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Recent Advances and Future Prospects in Gene Delivery Platforms for Glioblastoma
A Review

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IMPORTANCE Glioblastoma (GBM) remains a disease with high mortality despite treatment with maximal safe surgical resection, radiation, and temozolomide. Standard of care for GBM has remained unchanged for decades. Gene therapies using both viral and nonviral platforms have the potential to significantly improve outcomes but have faced challenges related to delivery, intratumoral heterogeneity, and immunologic limitations.

OBSERVATIONS This review discusses the most recent progress in genetic therapies for gliomas, with a focus on clinical platforms and those nearing clinical translation. Oncolytic and nonlytic replicating viral therapies remain the most clinically advanced, with adeno-associated virus and nonviral systems approaching clinical translation. Treatment efficacy has been variable across trials, platforms, and payloads. Oncolytic herpes simplex virus (G47Δ, teserpaturev) has been conditionally approved for the treatment of gliomas in Japan, while next-generation constructs show a potential association between immunologic activation and survival. Common challenges include the verification and quantification of delivery efficiency, the balance between antivector and antitumor immunity, appropriately designed clinical trials, relevant modeling of vector behavior in the preclinical setting, regulatory clearance, and manufacturing.

CONCLUSIONS AND RELEVANCE Understanding the potential and limitations of gene therapies for gliomas is vital in navigating the path forward for these promising therapies. The future of these treatments will focus on improved clinical trial design with rapid confirmation and tracking of delivery, biomarker feedback and assay optimization, and combination strategies. A focus on delivery platform development can compress iteration cycles and lead to therapeutic benefit for patients with gliomas and other solid tumors.

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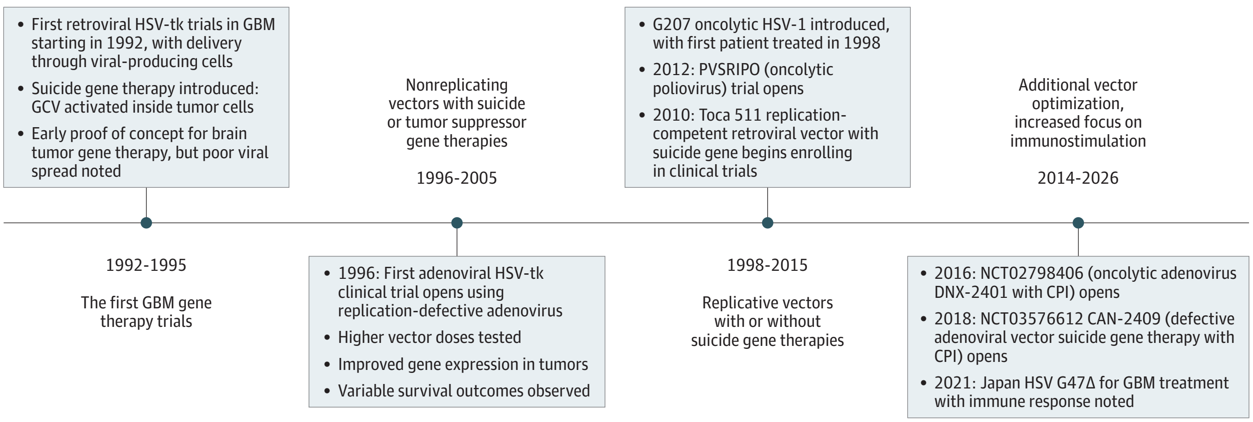
Advances in molecular biology continue to provide greater insight into the underpinnings of disease, creating further opportunities for direct perturbation of cellular systems using gene therapies, encompassing viral vectors, nucleic acid–based interventions, and gene-editing techniques. While lagging behind other treatments, such as small molecules and monoclonal antibodies in their current clinical impact, gene therapies, defined as the use of genetic material to treat or prevent disease, have entered the clinical setting for specific indications, including monogenic diseases and certain malignant neoplasms.[1] Novel genome-editing tools and advances in synthetic biology are increasing the potential options for therapeutic payloads and promise to further expand the indications for gene therapy.[2,3] These indications include both neoplastic and nonneoplastic pathologies.[4,5]

Among the target indications for gene therapies is glioblastoma (GBM), the most common and aggressive primary central nervous system malignant neoplasm, with an incidence of approximately 3 to 4 cases per 100 000 persons.[6] The unmet clinical need in GBM is high. Even with standard-of-care treatment with surgery, chemotherapy, and radiation, median survival is 12 to 17 months.[7] Given the sensitive location of these tumors, patient quality of life can also be significantly reduced, even in the setting of extended survival.[8] Standard-of-care treatment for GBM has continued without significant advancement for more than 2 decades.[9]

Despite the lack of a breakthrough therapy thus far, gene delivery represents a promising avenue of treatment for patients with GBM. Research in gene therapy for GBM first began in the 1990s with promising preclinical work in herpes simplex virus by Martuza et al,[10] followed by clinical trials in the late 1990s investigating suicide gene therapy using the implantation of murine retroviral producer cells producing retroviruses carrying the herpes simplex virus thymidine kinase (**Figure 1**).[11] Since these early studies, there has also been an increase in the use of these agents to stimulate an antitumor immune response rather than direct tumor cell killing (Figure 1). Challenges to the translation of promising preclinical results include tumor heterogeneity, therapeutic delivery, the immunosuppressive nature of the GBM tumor microenvironment (TME), and vector limitations. This review appraises recent progress achieved from bench

Figure 1. Timeline of the Evolution of Gene Delivery Platforms for Glioblastoma (GBM)



The evolution of gene delivery platforms for GBM is shown, divided into 4 historical epochs: (1) the initial clinical trials using replication-defective retroviruses carrying genes for prodrug-activating enzymes; (2) additional replication-defective vectors carrying suicide gene therapies; (3) the use of replication-competent vectors with or without suicide gene therapies; and (4) the increased focus on immunostimulation and additional vector optimization. CPI indicates checkpoint inhibitor; GCV, ganciclovir; HSV, herpes simplex virus; HSV-1, type 1 HSV; HSV-TK, HSV thymidine kinase; PVSRIPO, poliovirus-rhinovirus chimera.

to bedside in viral and other genetic therapies for GBM. We synthesize key mechanistic insights, highlight laboratory innovations, and summarize the current clinical trial landscape (eTable in the Supplement). We also provide a perspective on how these therapies will shape the next generation of GBM treatments and implications for the treatment of other cancer types.

Viral Platforms

Nonreplicative Adenovirus

Nonreplicative adenoviral vectors are nonenveloped double-stranded DNA viruses. They are frequently rendered replication incompetent through the deletion of early viral regions (E1 and E3), offer a packaging capacity of 7 to 8 kilobases (kb), and can result in robust but self-limited transgene expression.

Primarily delivered through intracranial injection, clinical efforts have evaluated the delivery of immunomodulatory or cytotoxic transgenes. One such approach used intracranial delivery of a nonreplicative adenoviral vector to encode interleukin 12 in a regulatable manner reliant on the oral activator veledimex. In a phase 1 trial involving 31 patients with recurrent high-grade glioma, this demonstrated increased lymphocyte infiltration and an acceptable safety profile.[12] Similarly, adenoviral delivery of herpes simplex virus (HSV) thymidine kinase (Ad-TK) followed by ganciclovir, a suicide gene approach, demonstrated a favorable safety profile in 2 phase 2 studies in patients with high-grade glioma with signs of clinical activity (one being a randomized study in the recurrent setting including intravenous and intra-arterial delivery[13] and the other being in newly diagnosed patients with direct injection at the time of surgery matched to control participants at another institution[14]). This included modest improvements in progression-free and overall survival when combined with standard-of-care treatment.[13,14]

The safety of this approach and likely antigen release with suicide gene–mediated cell death prompted the investigation of Ad-TK codelivery with a localized immunotherapy in the form of Ad-Flt3L.[15,16] This was designed to pair immunogenic cell death with Flt3L expression, which recruits dendritic cells and activates immune priming, with the goal of converting the resection cavity into an in situ vaccination. In a phase 1 trial of 18 patients, intracavitary coadministration of Ad-TK and Ad-Flt3L followed by valacyclovir was well tolerated and modestly surpassed 12- and 24- month survival rates relative to historical benchmarks. Treatment correlated with dendritic cell and $CD8^+$ T-cell infiltration.[17] Similarly, Wen et al[18] demonstrated in a phase 1b trial (n = 41) of newly diagnosed patients with GBM that intratumoral CAN-2409 (replication-deficient adenovirus expressing HSV TK suicide gene) plus valacyclovir combined with chemoradiation and adjuvant nivolumab was well tolerated. They observed a median survival of 30.6 months in patients with gross total resection of O6-methylguanine-DNA methyltransferase (MGMT)–methylated tumors, with evidence of immune activation correlating with survival.[18]

Conditionally Replicative Adenovirus

Conditionally replicative adenoviruses (CRAds) are nonenveloped double-stranded DNA viruses that exploit tumor-specific vulnerabilities to target virus replication to tumor cells. They have a larger genome of approximately 36 kb. The most clinically advanced CRAd is DNX-2401, which contains 2 genetic modifications to enhance tumor selectivity: a 24–base pair deletion in the adenovirus early region 1A (*E1A*) gene and an arginine-glycine–aspartic acid (RGD) motif inserted into the fiber H-loop.[19] The deletion in *E1A* prevents binding to the retinoblastoma (Rb) protein, which limits viral replication to cells with abnormal Rb function (common in GBM tumors). The RGD motif acts further upstream during viral entry into the host cell by facilitating entry through integrin binding. Now used in multiple settings and combinations, DNX-2401's first use in humans was a phase 1 window-of-opportunity study, published in 2018,[20] in 37 patients with recurrent GBM. Group A was a dose-escalation cohort with single injection of virus. This was followed by

Table. Summary of Gene Therapy Vectors for Glioma Treatment

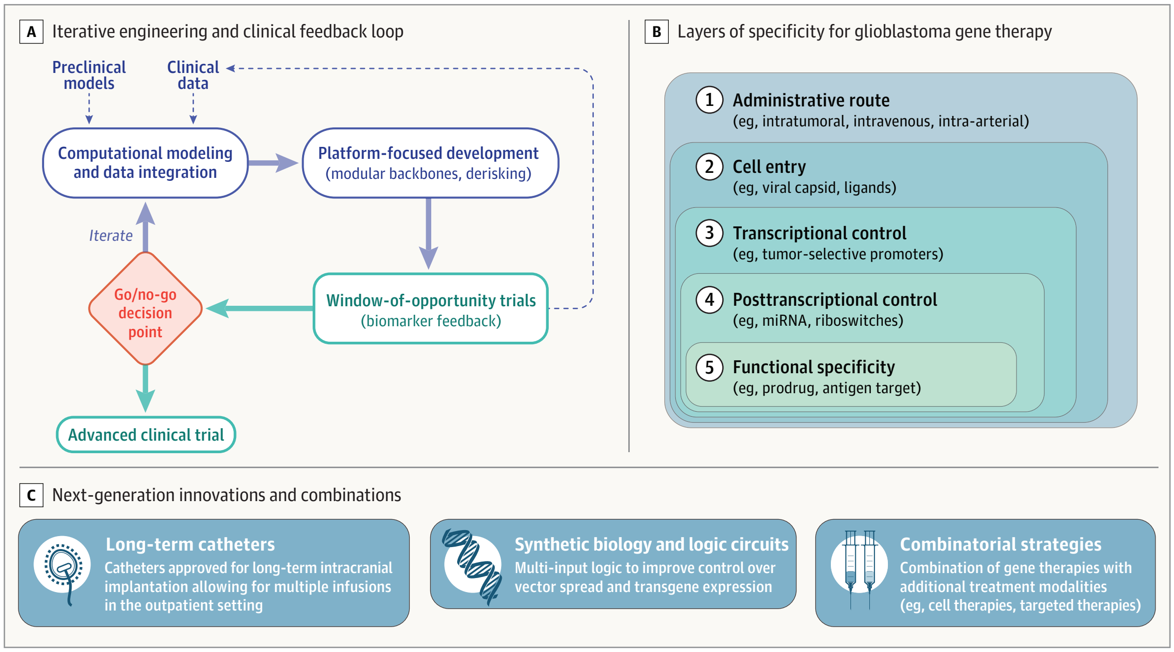
Vector	Size	Payload	Current stage for GBM	Advantages	Disadvantages
Adenovirus[12,17-21]	80-100 nm; genome, 26-45 kb	dsDNA	Phase 2 trials ongoing	Well studied and has wealth of historical data; larger genome with increased payload capacity	Systemic delivery may lead to hepatic sequestration and toxic effects; highly immunogenic and can trigger immune response, leading to rapid viral clearance
Oncolytic HSV[22-25,58]	150-225 nm; genome, 152 kb	dsDNA	Conditional approval of G47Δ in Japan	Large genome size, allowing for increased payload capacity; replication can be controlled with delivery of antiviral drugs; naturally neurotropic and can be engineered for further targeting specificity	IV delivery complicated by immune response and adherence to endothelium; immunogenic and can trigger antiviral immune response and clearance
RRV[26,28]	80-100 nm; genome, 7-10 kb	ssRNA	Phase 3 trial completed (Toca 511); other phase 1/2 trials ongoing	Stably integrates into cell genome and does not kill tumor cells as it replicates; strong clinical safety profile; poorly immunogenic with potential for greater persistence	Only infects actively dividing cells; potential risk for insertional mutagenesis (although reassuring profile in clinical trials)
Poliovirus[29,30,53]	25-30 nm; genome, 7.5 kb	ssRNA	Phase 2 trials ongoing	Naturally neuroinvasive, and CD155/Necl-5 receptor is overexpressed in tumor cells; sublethal infection of antigen-presenting cells can aid with generating antitumor immune response	Immunogenic, with potential for antiviral immune response and clearance
Measles virus[31,32,54,55]	100-300 nm; genome, 15.9 kb	ssRNA	Phase 1/2 trials ongoing	CD46 is overexpressed in tumor cells; potential bystander tumor cell killing effect	Limited ability to insert genetic payloads; systemic delivery limited by immunologic clearance
AAV[33-36]	20-25 nm; genome, 4.7 kb	ssDNA	Upcoming phase 1 trial	Several engineered variations have the ability to cross the BBB; widely used in preclinical and translational research	Smaller payload capacity compared with other viral vectors; systemic delivery has been associated with significant toxic effects; concerns about biodistribution and immunogenicity, depending on the capsid
VLP[37,38]	20-200 nm	RNA or DNA	Preclinical studies ongoing	Can be engineered to increase payload capacity and improve targeting	Biodistribution and therapeutic efficacy continue to be studied
LNP[39-41,56,57]	20-1000 nm	RNA or DNA	Phase 1 trial ongoing	Can be engineered via surface modifications to improve targeting specificity	Difficult to manufacture at scale; need for accurate in vivo tracking strategies; concerns about biodistribution
Nanocarriers[42-46]	1-1000 nm	RNA or DNA	Phase 1 trials ongoing	Can be engineered to improve targeting and BBB permeability; wide range of sizes, allowing for variation in payload capacity	Safety and biodistribution not fully understood; some materials have biocompatibility, scalability, and production cost concerns
AuNP[47,49,51,52]	1-200 nm	RNA or DNA	Phase 1 trial ongoing	Can be engineered to improve stability, biocompatibility, and BBB permeability; can be coupled with photothermal therapy; allows for concurrent imaging and diagnostic use	Safety profile not fully understood; requires modifications for accurate targeting of tumor cells; production process associated with environmental and biodegradability concerns

Abbreviations: AAV, adeno-associated virus; AuNP, gold nanoparticle; BBB, blood-brain barrier; dsDNA, double-stranded DNA; GBM, glioblastoma; HSV, herpes simplex virus; IV, intravenous; kb, kilobases; LNP, lipid nanoparticle; RRV, retroviral replicating vector; ssDNA, single-stranded DNA; VLP, viruslike particle.

11 patients in group B who received injection via implanted catheter and then received resection of their tumor 14 days later. Group A demonstrated that 20% of patients survived longer than 3 years, and the median overall survival was 13 months. Resected tumors from Group B revealed viral protein in 6 of 11 cases, accompanied by some evidence of tumor cell death, $CD8^+$ T-cell infiltration, and upregulated inflammatory genes. These data prompted combination trials of DNX-2401 with pembrolizumab in a multicenter phase 1/2 evaluation with dose-escalation and dose-expansion phases. This trial, where patients received a single dose of intratumoral DNX-2401 followed by pembrolizumab, again demonstrated a favorable safety profile as well as a 52.7% overall survival rate at 12 months and a 10.4% objective response rate.[19] Interestingly, objective responders tended to have tumors with an intermediate immune-cell profile marked by a high level of programmed cell death protein 1 (PD-1) but relatively low expression of other checkpoint molecules; this subtype was associated with a higher likelihood of clinical benefit and longer survival, implicating the immune system in treatment outcomes.[19]

An additional CRAd, CRAd-S-pk7, is engineered with a survivin promoter to drive *E1A* expression, thus making viral replication more tumor specific. It also involves a modification to the fiber, adding a polylysine (pk7) motif, which aids with targeting viral infection to tumor cells by facilitating binding to heparan sulfate proteoglycans. Interestingly, CRAd-S-pk7 was studied in combination with neural stem cells as a delivery vehicle in a first-in-human, phase 1 clinical trial involving 12 patients with newly diagnosed high-grade glioma. The therapy was injected by freehand in up to 10 sites in a tumor cav-

Figure 2. Schematic of the Framework for Next-Generation Glioblastoma Gene Therapies



A, Schematic demonstrating the closed-loop iterative engineering and clinical testing development cycle for novel glioblastoma gene therapies. Rapid development cycles with appropriate preclinical modeling lead to appropriately designed clinical trials with clear go vs no-go decision points and patient data that then inform next-generation vector design and payload choice. B, Illustration showing multiple layers of specificity for glioblastoma gene therapy targeting and control of therapeutic expression. C, Overview of selected next-generation strategies to overcome challenges in delivery and therapeutic efficacy, including implanted catheters, logic gating, and combinations with other targeted therapies. miRNA indicates micro RNA.

ity after resection and was found to be safe and well tolerated. Median overall survival was 18.4 months. Posttreatment TME analysis demonstrated a significant increase in CD8$^+$ T-cell infiltration. Interestingly, in 8 of 12 patients who received additional brain sampling during re-resection or autopsy, there was no evidence of viral presence (E1A and hexon) or neural stem cell presence (v-myc DNA).[21] In summary, the clinical evidence around advanced adenoviral vectors suggests stimulation of the immune system and modification of the TME without a significant clinical response.

Oncolytic HSV

HSV is an enveloped double-stranded DNA virus with a large genome of approximately 152 kb. The well-defined nature of the HSV genome allows for the deletion of genes responsible for wild-type neurovirulence and the insertion of tumor-selective promoters as well as therapeutic payloads. The first US Food and Drug Administration–approved oncolytic HSV was talimogene laherparepvec (T-VEC), which also carries granulocyte-macrophage colony-stimulating factor, for advanced melanoma treatment via local injection.[22] In GBM, G47Δ has received the first regulatory approval. In a single-arm phase 2 study of G47Δ in patients with recurrent GBM, Todo et al[23] demonstrated a significant benefit with a 1-year survival rate of 84.2% after G47Δ initiation, meeting a prespecified study end point. A defining and relatively unique feature of their study was repeated intratumoral injections—up to 6 per patient.[23] This resulted in a conditional time-limited approval in Japan. Interestingly, preclinical studies of G47Δ have implicated both direct oncolytic activity and induction of specific antitumor immunity as contributing to treatment efficacy.[24] Histological analysis of samples from patients who received an injection of G47Δ has also demonstrated infiltration of CD4$^+$ and CD8$^+$ T cells, seemingly correlating with preclinical findings.[23] An additional notable finding of the study was the impressive survival benefit without a drastic change in imaging findings, as the overall response rate was only 5.3%, which Todo and colleagues ascribed to confounding by infiltrating tumor lymphocytes.[23]

Additional HSV constructs for GBM remain in clinical trials. This includes CAN-3110 (rQNestin34.5v.2) which includes the neurovirulence gene *ICP34.5* but uses a nestin promoter to limit replication to GBM cells. In a first-in-human phase 1 study of CAN-3110 involving 41 patients with recurrent GBM, no dose-limiting toxic effects were seen. Interestingly, patient survival was associated with changes in infiltrating tumor T-cell counts, specific T-cell clonotypes, and increased tumor transcriptomic immune activation.[25] Both pretreatment and posttreatment HSV type 1 serology findings were also associated with prolonged survival as well as viral clearance from the tumor. These findings suggest an association between an antiviral, and possibly antitumor, immune response and treatment efficacy.[25]

Retroviral Replicating Vectors

Retroviral replicating vectors (RRVs) have a long history in cancer gene therapy. Originating from the murine leukemia virus, RRVs have a positive-sense single-stranded RNA genome and are unique among cancer gene therapy vectors in that they replicate and do not lyse cells as part of their life cycle.[26] As such, the primary therapeutic benefit from RRVs is generated from the payload delivered rather than the virus itself. The most clinically advanced generation of RRV pursued a suicide gene strategy by delivering yeast cytosine deaminase followed by a prodrug (5-fluorocytosine) that is converted to the chemotherapeutic 5-fluorouracil within tumor cells. This led to localized chemotherapeutic activity and secondary immune activation in preclinical models.

Strong results led to clinical translation in the form of vocimagene amiretrorepvec (Toca 511) in 2010. Multicenter dose-escalation phase 1 clinical trials with Toca 511 in recurrent high-grade glioma evaluated 3 different dosing strategies: intratumoral injection, resection and injection, and intravenous delivery, with good safety, tumor selectivity, and median survival of 11 to 14 months.[26] There was also some evidence of long-term survivorship in a subset of patients. As such, Toca 511 then advanced to a randomized phase 3 trial in patients with recurrent high-grade glioma. This trial failed to meet its primary end points, in part likely due to the low median number of prodrug cycles received by patients. Post hoc analysis of the phase 3 trial demonstrated some possible benefit in patients with second recurrence and anaplastic astrocytomas.[27] RRVs remain in clinical development with Toca 511, now DB107, in a phase 1 trial and KB-516, which directly targets the immune system, approaching the clinic.[26,28]

Other Oncolytic Viruses

Poliovirus is a small, nonenveloped RNA virus. The most clinically advanced construct is poliovirus-rhinovirus chimera (PVSRIPO), a live attenuated vaccine strain engineered to replace its internal ribosomal entry site with that of human rhinovirus type 2. In a phase 1 dose-escalation study enrolling 61 patients, PVSRIPO demonstrated an acceptable safety profile, with less than 20% of patients experiencing adverse events of grade 3 or higher.[29] The most significant finding of this trial was the distinctive survival pattern: overall survival among patients who received PVSRIPO reached a plateau beginning at 24 months, with the overall survival rate being 21% at both 24 and 36 months, whereas overall survival in the historical control group continued to decline from 14% at 24 months to 4% at 36 months. The tail of overall survival observed in the treatment group may represent an antitumor immune response. Preclinical studies have demonstrated complementary action between the release of antigens during virus-mediated tumor cell death and type 1 interferon activation as a result of sublethal infection of antigen-presenting cells in the TME.[30] Notably, 2 patients survived longer than 69 months following their PVSRIPO infusion. PVSRIPO was granted breakthrough designation by the US Food and Drug Administration in May 2016, and phase 2 trials are ongoing.

The measles virus (MV) is an enveloped single-stranded RNA virus. The attenuated Edmonston MV vaccine lineage has served as the backbone for therapeutic development, with several preclinical studies demonstrating the efficacy of engineered derivatives expressing the human carcinoembryonic antigen (MV-CEA) or sodium iodide symporter (MV-NIS) against GBM models.[31] The phase 1 first-in-human trial of this construct involved 22 patients with recurrent GBM, 9 of whom received tumor resection followed by intracavitary injection of MV-CEA and the remaining 13 of whom received an intratumor dose preoperatively and a second, postresection intracavitary injection 5 days later.[32] Treatment was safe, with no dose-limiting toxic effects at the maximum tested dose. Median overall survival was 11.6 months, with a 1-year survival rate of 45.5%. Of note, the authors determined that posttreatment levels of viral genomes and CD8+ cell infiltration in the studied GBMs could be predicted by baseline expression of interferon-stimulated genes. This facilitated the development of a predictive algorithm for viral infection and spread.

Adeno-Associated Viruses

Adeno-associated viruses (AAVs) are nonenveloped single-stranded DNA viruses. These viruses have genomes of approximately 4.7 kb. Several variants, including AAV9, AAV2, and AAVrh10, have demonstrated neurotropism.[33,34] Recombinant AAV vectors are replication incompetent. A recently published preclinical study demonstrated that AAV-delivered CX-C motif ligand 9 (CXCL9) therapy can increase GBM responsiveness to anti–PD-1 treatment.[35] The investigators demonstrated that AAV-CXCL9 constructs improved recruitment of $CD8^+$ lymphocytes to the TME and improved survival in both the GL261 and KR158 glioma models, with 50% and 25% of animals demonstrating long-term survival, respectively. Another construct, NXL-004, an AAV delivering the neural transcription factor NeuroD1, has also been shown to be efficacious in reducing GBM tumor volume in a mouse model.[36] Despite these promising features, an AAV has not yet been investigated in clinical trials for GBM, although a study involving NXL-004 has been prospectively registered (Chinese Clinical Trial Register identifier: ChiCTR2400080362) to evaluate the safety in humans and another study investigating TGX-007 is on the horizon.

Nonviral Delivery

Viruslike Particles

Viruslike particles (VLPs) are nanostructures composed of viral capsid proteins lacking viral genomic material. Traditionally synthesized from retroviral components, VLPs were generated with the hope of offering the advantages of viral vectors, including efficient cellular entry and gene transfer with increased versatility, but they have had stability and manufacturing issues. The JC polyomavirus (JCPyV), a neurotropic virus that naturally infects glial cells and oligodendrocytes, is one vector that has been adapted into a VLP-based gene delivery platform for GBM. In a preclinical study, JCPyV VLPs successfully transduced U87 glioma cells and significantly prolonged survival when loaded with the HSV thymidine kinase suicide gene and administered intratumorally with the ganciclovir prodrug.[37] No clinical trials to date have evaluated VLP-mediated gene delivery in patients with GBM. Translation of VLPs remains limited by challenges with delivery, immunogenicity, and tumor specificity.[38]

Lipid-Nanoparticles

Lipid nanoparticles (LNPs) are artificial, spherical microvesicles composed of ionizable lipids, phospholipids, cholesterol, or polyethylene glycol conjugates. LNPs can range from 20 to 1000 nm in diameter.[39,40] Surface modification strategies have enhanced LNP targeting specificity for glioma cells through conjugation with ligands, such as transferrin, lactoferrin, or other peptides, recognized by overexpressed glioma receptors. One construct is CVGBM, constituting an LNP encapsulating messenger RNA (mRNA) encoding a multiepitope vaccine for GBM. A phase 1 trial is now underway to evaluate the efficacy of CVGBM in patients with MGMT-unmethylated GBM (ClinicalTrials.gov identifier: NCT05938387). Challenges for the translation of lipid nanoparticles to clinical practice include manufacturing multitargeted LNPs at scale and performing accurate in vivo tracking of LNP delivery.[41]

Nanocarriers

Nanocarriers are synthetic nanoscale delivery systems engineered from a wide range of materials, including proteins, lipids, or polymeric nanoparticles. In preclinical studies, these platforms have demonstrated efficient tumor uptake, gene silencing, and survival outcomes. In one study, human H-ferritin was leveraged to create nanocarriers with the goal of disassembly in endosomes and lysosomal escape of small interfering RNA (siRNA) cargo.[42] These ferritin-based nanocarriers demonstrated effective crossing of the blood-brain barrier and targeted delivery to glioma cells, resulting in improved suppression of GBM progression-associated gene expression. In another approach, IRF5/IKKβ mRNA-loaded polymeric nanoparticles were used to reprogram tumor-associated macrophages in a platelet-derived growth factor β–driven glioma mouse model.[43] Another example of nanocarriers includes high-density lipoprotein (HDL)–based nanodiscs, which use endogenous lipoprotein transport pathways to deliver cargo to target cells. Preclinical models have demonstrated the ability of HDL-based nanodiscs to deliver combination chemotherapeutics and immunotherapeutics (CpG and docetaxel), with promising preclinical findings.[44] Similarly, synthetic protein nanoparticles (cross-linked human serum albumin) have demonstrated the ability to penetrate the blood-brain barrier and accumulate in tumors. They have been shown to deliver a range of payloads, including radiosensitizers and immunotherapies in preclinical models.[45,46] These studies highlight the versatility of nanocarriers in GBM therapy. While several nanocarrier platforms are in early-phase clinical trials, long-term safety, biodistribution, and scalable manufacturing remain active areas of investigation.

Gold Nanoparticles

Gold nanoparticles (AuNPs) are inorganic nanostructures typically ranging from 1 to 200 nm in diameter, with modifiable surface chemistry that enables conjugation with peptides, targeting ligands, or nucleic acids.[47,48] In GBM treatment, AuNPs have been explored as vehicles for siRNA and micro RNA delivery, with preclinical studies demonstrating successful GBM targeting, gene silencing, and tumor growth inhibition.[49,50] The first phase 0 in-human trial (ClinicalTrials.gov identifier: NCT03020017) evaluated spherical nucleic acid constructs (NU-0129) composed of AuNPs functionalized with radially oriented siRNA oligonucleotides targeting Bcl2L12.[51] In this study, patients with recurrent GBM received intravenous NU-0129 prior to tumor resection. The therapy was well tolerated, and analysis of resected tumor tissue confirmed gold enrichment within macrophages, endothelial cells, and tumor cells. Importantly, NU-0129 uptake was also correlated with reduced Bcl2L12 protein expression. A remaining challenge for AuNPs is the need to elucidate biosafety, as toxicity is highly dependent on particle size, shape, and surface chemistry, which can either mitigate or exacerbate adverse effects.[52]

A summary[12,17-26,28-47,49,51-58] of GBM gene therapy vectors can be found in the **Table**.

Challenges and Open Questions

Despite the promise of the aforementioned gene therapies, a number of challenges exist limiting their widespread use for the treatment of GBM. Primary among these is construct delivery to the desired cell or tissue in patients. While delivery has been practically measured in some clinical trials, methods and measures of successful delivery vary significantly between trials. Biodistribution and tumor penetrance and delivery are critical components that can significantly impact therapeutic efficacy. More stringent measures of delivery should be used with orthogonal modalities confirming and quantifying construct presence in the tumor and elsewhere, as able, as a primary outcome that is reported and used to contextualize results. This is especially pertinent given the literature around antivector immunologic reactions that can limit delivery and blunt desired antitumor responses. This is highlighted by Webb et al,[59] who demonstrated that an oncolytic vesicular stomatitis virus expressing interferon β generated a significant antiviral response that antagonized the impact of anti–programmed death-ligand 1 (PD-L1) treatment in a hepatocellular cancer model. In addition, rapid clearance of oncolytic viruses can of course limit viral spread and therapeutic payload delivery, thus limiting efficacy.[60,61] Various routes of delivery, including intravenous, intratumoral injection, intratumoral implants, intra-arterial, intrathecal, convection-enhanced delivery, intranasal, cell carriers, and others (**Figure 2**), should also be carefully assessed and measured in the preclinical setting while considering unique features of the vector being tested.[62,63] As such, future trials should carefully validate delivery route as a key consideration in determining the efficacy of their intervention and also consider a comparison between multiple routes of delivery.[63]

Considerations around delivery route also highlight the significant gap between preclinical models and human patients. Many preclinical models involve tumor cell lines that have been implanted into murine brains. These are often homogeneous and grow as well-demarcated, spherical shapes that do not recapitulate the barriers to vector delivery, such as fibrosis or necrosis, that a tumor would have in a human.[64,65] In addition, the blood-brain barrier and vasculature of murine GBM models likely do not accurately represent that of tumors in humans. The literature has demonstrated significant differences in degree of blood-brain barrier disruption between murine models and tumors in humans, differences in transporter protein expression, and other molecular incongruencies.[66-70] These differences limit preclinical insights into therapeutic delivery strategies and highlight the need for additional models. The lack of biomimetic models for preclinical testing of delivery also highlights the importance of window-of-opportunity trial designs and tissue sampling in GBM gene therapy clinical trials, which can also provide insight into treatment response. This is highlighted in recent work by Ling et al,[25,71] where tissue sampling and serial multiomics have shed light on underlying immunologic changes in response to treatment with CAN-3110 and on possible predictors of clinical response.

Human tissue data and biomimetic models are especially important in modeling the GBM TME, which poses multiple barriers to gene therapy delivery and efficacy in these tumors. Physical barriers to gene delivery include the dense extracellular matrix, necrotic regions, and cerebrospinal fluid dynamics. Abnormal tumor vasculature and the blood-brain barrier can similarly limit vector delivery through intravenous or intra-arterial delivery.[63] As previously mentioned, immune responses against gene therapy vectors can limit vector distribution and hinder efficient gene delivery. In addition, immunosuppressive pathways and cell populations in the GBM TME, including myeloid cell populations, can suppress antitumor immunity and the efficacy of immunostimulatory payloads.[72] Novel therapeutic strategies aimed at targeting and reprogramming immunosuppressive myeloid populations, such as those described by Zhang et al[43] and Montoya et al,[73] represent promising paths to overcome the limitations of the immunosuppressive GBM TME. Recognition and further understanding of these barriers will play a key role in next-generation vector design and use.

Finally, manufacturing and time to the clinic remain challenges across cell and gene therapies. Reaching good manufacturing practices–scale production, large-scale manufacturing batches, and release testing are costly and time-consuming. In addition, regulatory approval to begin human trials can significantly extend timelines. A focus on platform development and modular practices can help reduce manufacturing times and aid with establishing regulatory pathways to the clinic. Given the limitations of preclinical systems, human data are a critical component of viable therapeutic development.

Future Directions

Upcoming studies of GBM gene therapies may benefit from prioritizing platform development, rapid human data generation and iteration, and the application of modern molecular methods (Figure 2). Platform-focused development with the validation of vector backbones in achieving adequate delivery and safety may then facilitate the rapid testing and iteration of therapeutic payloads. Human tissue–based preclinical models should be used when appropriate. Well-designed clinical trials with rapid biomarker feedback and strict go vs no-go decision points can derisk delivery and facilitate treatment iteration and improvement. Window-of-opportunity trial designs and frequent tissue sampling can provide valuable insight into vector distribution, persistence, and expected payload-related changes. Artificial intelligence similarly has the opportunity to integrate preclinical and human data to inform design changes and iteration of glioma gene therapies.

Additional innovation in gene therapies may include incremental advances, like catheters approved for long-term implantation that could allow multiple intracranial infusions of a vector in the outpatient setting; strategies combining genetic therapies with other innovative therapies, such as systemic cellular therapies; or the innovative application of synthetic biology techniques.[74] For example, GBM gene therapies can prime the TME for combination treatment with therapies like chimeric antigen receptor (CAR) T-cell therapy. Gene therapy vectors could be used for the delivery of CAR T-cell epitopes or to establish chemokine gradients.[75,76] Additional payloads, such as CRISPR-Cas systems, are also currently being evaluated preclinically for knockout of immunosuppressive targets to further sensitize tumors to immunotherapies (eg, CD47 and PD-L1).[77,78] Of note, the more artificial the payload is, the greater the risk is for immunologic limitations to vector distribution and payload expression as well as safety considerations. Genome editing is also being used to modify cell therapies ex vivo (ClinicalTrials.gov identifier: NCT06815029), where transforming growth factor β receptor 2 (TGFBR2) knockout is used to arm CAR T cells against the immunosuppressive GBM TME.[79] Alternatively, next-generation vectors may also use synthetic biology techniques, including multi-input logic to improve control over vector spread and transgene expression, potentially improving the safety and efficacy of these therapies. Leveraging multiple layers of specificity, including delivery modality, mediators of host cell entry, and transcriptional or translational control, may aid in improving therapeutic activity in target cell populations (Figure 2). This may allow for not only improved safety but also increased payload expression, thus widening the therapeutic window. These tools, in combination with more advanced modeling systems, may also allow for the development of vectors that can overcome delivery barriers, such as necrosis and fibrosis in a tumor. Together, the future of GBM gene therapies will likely involve data-driven, controllable platforms that leverage human data to inform rapid iteration and improvements to generate novel therapies for patients.

ARTICLE INFORMATION

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